

Automobile Valuation, Resale & Depreciation Analytics

Comprehensive exploratory analysis across all 10 scenarios covering depreciation dynamics, transmission premiums, and fleet metrics.

Dataset: cardata.csv (301 Records)

Metrics: Selling_Price, Present_Price, Kms_Driven

Tech: Pandas, NumPy, Matplotlib

1. Scenario Breakdown (All 10 Tasks)

Scenario / Task	Method / Distribution	Core Finding / Key Metric
S1: Ingestion & Stats	Imputed nulls; NumPy stats	Min: 0.10L, Max: 35.00L, Mean: 4.66 Lakhs.
S2: Price Trend	First 10 rows line graph	Sample dispersion from 2.85L to 7.25 Lakhs.
S3: Expensive (>10L)	Fuel filtering + bar chart	Diesel represents 82.4% of luxury cars (28 vs 6 Petrol).
S4: Fuel Pie Chart	Value counts + pie chart	Petrol: 79.4% (239) Diesel: 19.9% (60) CNG: 2.
S5: Price Scatter	Present vs Selling price	Strong positive correlation ($r = 0.878$).
S6: Age Categories	New (>2015), Med (2010-14), Old (<2010)	Medium-aged cars dominate active inventory (70.8%).
S7: Mileage Histogram	Kms_Driven (25 bins)	Right-skewed; most cars clustered below 50,000 kms.
S8: Transmission Resale	Groupby Transmission mean	Automatic cars command a +139.7% premium (9.42L vs 3.93L).
S9: Seller Breakdown	Dealer vs Individual counts	Dealers account for 64.8% (195) vs Individuals 35.2% (106).
S10: Depreciation	Diff = Present - Selling	Avg Depreciation: 2.97L, Max: 59.85L. Older cars devalue rapidly.

2. Visual Representation

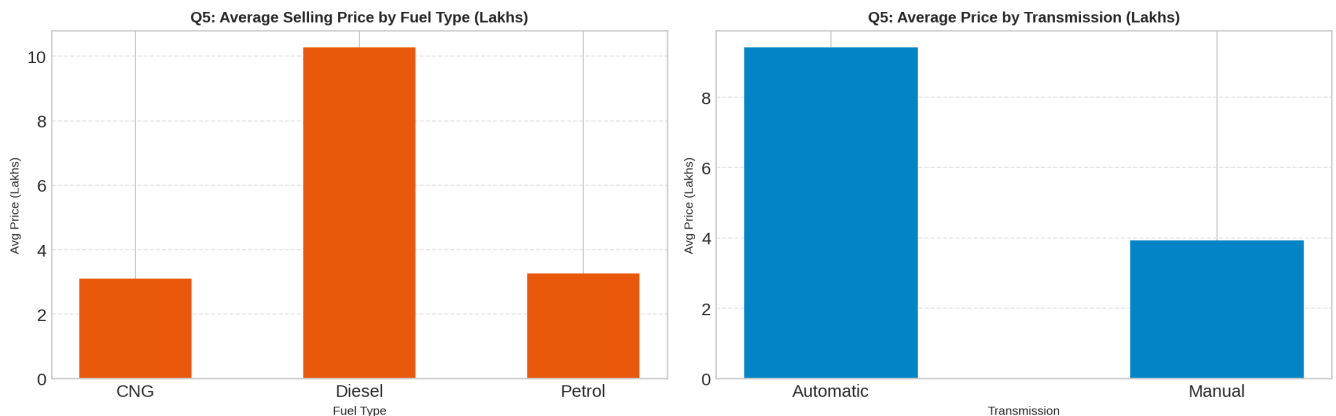


Figure 5.1: Average Selling Price by Fuel Type (Left) & by Transmission Type (Right).

3. Python Code Snippet

```
import os, pandas as pd, numpy as np, matplotlib.pyplot as plt
df = pd.read_csv("cardata.csv")
for c in ['Selling_Price', 'Present_Price', 'Kms_Driven']:
    df[c] = pd.to_numeric(df[c], errors='coerce').fillna(df[c].mean())
df['Price Difference'] = df['Present_Price'] - df['Selling_Price']
print(f"Automatic vs Manual Price: {df.groupby('Transmission')['Selling_Price'].mean().to_dict()}")
```